

SEMESTER END EXAMINATIONS – MAY 2023

Program : B.E. – Information Science and Engineering
Course Name : Machine Learning
Course Code : IS61

Semester : VI
Max. Marks : 100
Duration : 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Discuss the types of Machine Learning. CO1 (06)
 - Illustrate the process of machine learning with a neat diagram. CO1 (05)
 - The test data set is having 44 instances and below is the confusion matrix after testing a binary-class classification model. CO1 (09)

	Predicted		
Actual		P	N
	P	20	10
	N	4	10

Calculate Accuracy of the model and precision, recall of all target classes.

- Distinguish between supervise learning and reinforcement learning. CO1 (07)
 - Discuss the trade-off between bias and variance CO1 (04)
 - Explain the following terms: CO1 (09)
 - Over-fitting
 - ROC Curve
 - Unbalanced Dataset.

UNIT - II

- Identify the first splitting attribute for the decision tree by using the ID3 algorithm with the following dataset. CO2 (06)

Major	Experience	Tie	Hired
CS	programming	pretty	NO
CS	programming	pretty	NO
CS	management	pretty	YES
CS	management	ugly	YES
Business	programming	pretty	YES
Business	programming	Ugly	YES
Business	management	pretty	NO
Business	management	pretty	NO

- Explain the similarity and differences between linear and logistic regression models. CO2 (06)
- Illustrate the Linear Discriminant Analysis with an example. Suggest any one application where LDA can be used. CO2 (08)

4. a) A sample example of a regression equation to predict the glucose level given the age is as follows: CO2 (06)

SUBJECT	AGE X	GLUCOSE LEVEL Y1
1	43	99
2	21	65
3	25	79
4	42	75
5	57	87
6	59	81
7	55	?

Predict the Glucose Level for Subject 7 using Linear Regression Model.

- b) Describe ID3 algorithm. Interpret how is ID3 different from C4.5 algorithm. CO2 (06)
- c) For the given dataset, find the person whose weight = 57kg and Height =170cm is underweight or Normal using k Nearest Neighbor Algorithm. CO2 (08)

Height (cm)	Weight (kg)	Class
167	51	Underweight
182	62	Normal
176	69	Normal
173	64	Normal
172	65	Normal
174	56	Underweight
169	58	Normal
173	57	Normal
170	55	Normal
170	57	?

UNIT - III

5. a) Consider the following data given below, using KNN algorithm with value of $k = 3$, find the weight for data: Height= 5.5 and Age= 38. Provide a detailed step-by-step solution to the problem. CO3 (10)

Table 4: data

Height	Age	Weight
5	45	59
5.11	46	77
5.6	55	61
5.7	56	65
5.4	28	50
5.3	35	53
5.0	46	59
4.11	26	67
5.3	57	57
5.5	43	55

- b) Given the Fig.5(b-1) and Fig.5(b-2) below reflect on difference in approach for classification used Support Vector Machine(SVM). Provide detailed discussion on support vectors, hard & soft margin, kernel trick used for classification using SVM. CO3 (10)

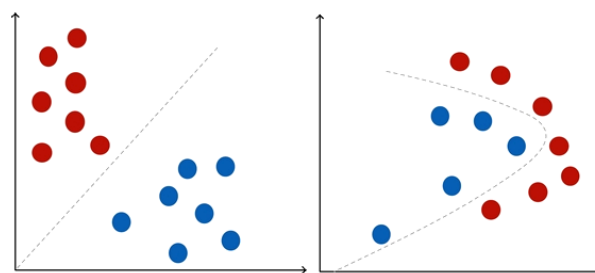


Fig.5(b-1)

Fig.5(b-2)

6. a) Reflect on Support Vector Machine(SVM) approach classification. What is the “kernel trick” and how is it useful? Give example scenarios. CO3 (08)
- b) With relevant mathematical equations discuss Naïve Bayes Classification. Consider the following hypothetical data in table 1 concerning student characteristics and whether or not each student should be hired. Use a Naive Bayes classifier to determine whether or not someone with *excellent attendance, excellent GPA, and some effort* should be hired. Provide a detailed step by step approach to solution. CO3 (08)

Table 1: student details

Name	GPA	Effort	Hirable
Sarah	poor	lots	Yes
Dana	average	some	No
Alex	average	some	No
Annie	average	lots	Yes
Emily	excellent	lots	Yes
Pete	excellent	lots	No
John	excellent	lots	No
Kathy	poor	some	No

- c) What happens when you increase or decrease the value of K in KNN? Illustrate using an example. CO3 (04)
- UNIT- IV**
7. a) Apply k-meansclustering algorithm with Euclidean distance measure to Cluster the following 8 data points A1(2,10), A2(2,5), A3(8,4), A4(5,8), A5(7,5), A6(6,4), A7(1,2), A8(4,9), Initial centroids are data points A1, A4 and A7. CO4 (10)
- b) Consider the two dimensional patterns (2, 1),(3, 5),(4, 3),(5, 6), (6, 7),(7, 8) and compute the principal component using PCA Algorithm. CO4 (10)
8. a) Using AGNES cluster the data points A(1,1), B(1.5,1.5), C(5,5), D(3,4), E(4,4), F(3,3.5) using Euclidean distance. Draw the Dendrogram and the Clustering Hierarchy in X-Y space. CO4 (10)
- b) Group the below dataset into two clusters using k-means algorithm. OB-2 and OB-6 are centroids of Cluster 1 and Cluster 2 respectively. Use Manhattan distance. CO4 (10)

Objects	X	Y	Z
OB-1	1	4	1
OB-2	1	2	2
OB-3	1	4	2
OB-4	2	1	2
OB-5	1	1	1
OB-6	2	4	2
OB-7	1	1	2
OB-8	2	1	1

UNIT - V

9. a) What is a Perceptron? Illustrate a Perceptron for AND function with a diagram. CO5 (06)
- b) Illustrate your understanding of Gradient Descent? Explain three different types of Gradient Descent with examples. CO5 (06)
- c) Write the 'Forward Pass Algorithm' used in MLP with any valid example. CO5 (08)
10. a) What do you understand by the term 'Linear Separability'? Illustrate a few graphs which depicts linear and Non-Linear Separability. CO5 (06)
- b) What do you understand by the term 'MLP(Multi-layer Perceptron)'? What are the limitations of Single Layer Perceptron which led to the evolution of MLP? CO5 (06)
- c) For the MLP given Fig.10(c), calculate the Total Error(E_{total}) after Forward Pass at Output Layer and calculate the modified weight of w_5 . CO5 (08)

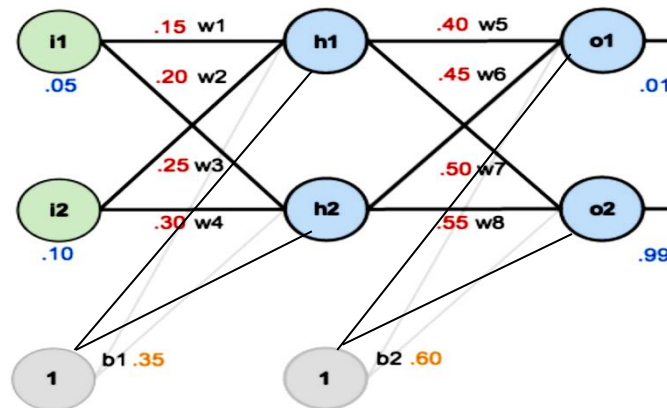


Fig.10(c)

Ensure that you show one 'epoch' (one forward pass and one backward pass).
